



Messages crashing PS4s

A number of PS4 users on Reddit and other social media are reporting a malicious message that crashes the console. <https://digit.in/nov18-77-1>



Your poop is full of plastic

A study conducted by Vienna researchers showed the presence of up to 9 different plastic types in faeces. <https://digit.in/nov18-77-2>



The OSIRIS-REx mission

Credit: NASA

observation campaign, OSIRIS-REx will study the chemistry, mineralogy and global properties of the asteroid using the on board instruments.

During this period, a number of practice maneuvers will be conducted, without actually gathering the sample. The two solar panels on board will be deployed in a Y shaped configuration. This is to minimise the accumulation of dust on the panels during the sample gathering process, and to minimise damage to the spacecraft in case it tips over and gets uncomfortably close to the asteroid. NASA has developed a touch-and-go sample gathering arm, which hangs below the spacecraft. If all goes well, this arm is the only part of the spacecraft that will touch the surface. After a few practice runs, the actual sample gathering will be attempted. This is a complicated process considering that the arm has a very small time window to actually gather the sample. Once deployed, the arm will make contact with the

surface of the asteroid, through accelerometers. A spring mechanism on the arm acts as a shock absorber, protecting the spacecraft from the jolt of the impact. A burst of nitrogen gas, will throw small particles of the regolith into the collector. After five seconds, the spacecraft will automatically increase its altitude, and check if enough material has been collected. If the sample is not of suf-

ficient quantity, then the spacecraft will attempt to gather it again. There is enough nitrogen gas on the arm for a total of three attempts. The sample gathering success of OSIRIS-REx is contingent on sufficient regolith to be available at the gathering site.

INTER AGENCY COLLABORATION

JAXA and NASA are actually working together on both the missions. Both the OSIRIS-REx and the Hayabusa 2 missions hope to return samples from asteroids formed very early during the creation of the solar system. As such, the asteroids can be considered time capsules, that will allow scientists to study the origins of the solar system and the Earth. To do this as accurately as possible, the scientists require as many samples, from as many diverse locations as possible. Either asteroid may be an outlier case, instead of the norm. The specific sites from which the samples are gathered may also have unusual properties. The idea is to ensure that the returned samples are representative of carbonaceous asteroids. Towards this goal, the samples from the two missions will be shared with the scientists from the other team. 10 percent of the Hayabusa 2 samples, and 0.5 percent of the OSIRIS-REx samples will be shared with the other team. The discrepancy is because OSIRIS-REx will be returning a much larger quantity of material than Hayabusa 2. Additionally, to ensure a deep collaboration, there are JAXA scientists in the OSIRIS-REx mission, and NASA scientists in the Hayabusa 2 mission.

At this point in time, these are the only two asteroid sample missions. There are two planned future missions to asteroids, neither of which are sample return missions. The Lucy mission will explore the trojan asteroids of Jupiter, a group of objects captured from the asteroid belt. The Psyche mission is to an m type asteroid of the same name. Lucy and Psyche are both missions planned by NASA as part of the Discovery Program. 



The Minerva rovers on Hayabusa 2

Credit: JAXA